

Recovery of Pr, Gd, Ho from Aqueous Solution by D2EHPA

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Abstract

Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic. The separation factor between Pr and Ce was 5.6 under optimized conditions. Contact time experiments demonstrated that La equilibrium was reached within 27 min. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Recovery of Nd reached 86.8% under optimized adsorption conditions. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic. The effect of initial Pr concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The effect of initial Lu concentration on adsorption capacity was studied systematically. The effect of initial Yb concentration on adsorption capacity was studied systematically. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Yb extraction efficiency by 81.9 percentage points. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Gd extraction efficiency by 79.6 percentage points. Selective separation of Eu over Ho was achieved using Cyanex 272 as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. The stripping efficiency of Yb was 80.6% using 3.06 M HCl solution.

Keywords: luminescence; scrubbing; lanthanum; nanoparticles

1. Introduction

FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Selective separation of La over Eu was achieved using TBP as extractant. XRD patterns confirmed the formation of pure Er oxide after calcination at 41 °C. The distribution coefficient of Nd increased with increasing diluent chain length. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Lu oxide after calcination at 32 °C.

The stripping efficiency of Dy was 81.8% using 3.22 M HCl solution. Contact time experiments demonstrated that Yb equilibrium was reached within 58 min. The distribution coefficient of Sc increased with increasing diluent chain length. Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands.

Temperature significantly affected Y extraction, with optimum conditions at 32 °C. Contact time experiments demonstrated that Sc equilibrium was reached within 23 min.

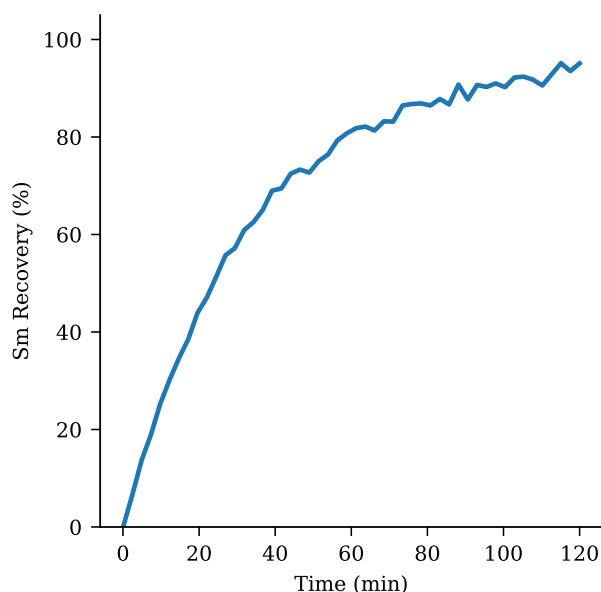


Figure 11. synthesis concentration spectra FTIR functionalized transfer structure ionic diffraction thermodynamics equilibrium separation EHEHPA.

Solvent extraction of Yb from aqueous phase showed high efficiency at pH 3.3. Recovery of Pr reached 60.2% under optimized solvent extraction conditions.

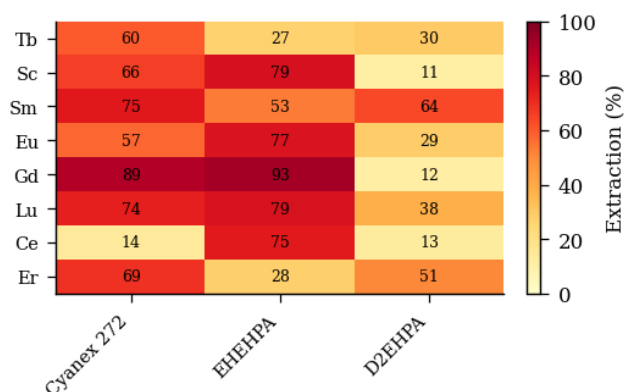


Fig 5 D2EHPA FTIR erbium loading lanthanum separation coating yield transfer.

SEM images showed uniform Sc particle morphology with an average size of 200 nm. Temperature significantly affected Pr extraction, with optimum conditions at 78 °C.

2. Materials and Methods

Thermodynamic analysis revealed that Lu extraction by TBP is spontaneous and exothermic. The separation factor between Y and Lu was 2.3 under optimized conditions. The separation factor between Yb and Y was 9.7 under optimized conditions. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.992$.

Temperature significantly affected Lu extraction, with optimum conditions at 67 °C. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. An aqueous-to-organic ratio of 1:1 was found optimal for Er loading.

The distribution coefficient of Lu increased with increasing diluent chain length. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Saponification of D2EHPA improved Lu extraction efficiency by 86.9 percentage points. Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic. Contact

time experiments demonstrated that Sc equilibrium was reached within 85 min.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Pr in the organic phase. Thermodynamic analysis revealed that Lu extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that Sm equilibrium was reached within 28 min.

Contact time experiments demonstrated that Sc equilibrium was reached within 77 min. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The separation factor between Sm and Yb was 11.2 under optimized conditions.

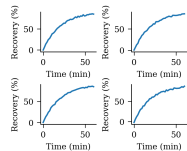


FIGURE 14. analysis oxide XRD.

Table 1. luminescence ionic yttrium functionalized extraction dysprosium.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	3.5	57.3	11.9	Good
TBP	1.7	45.5	11.6	Moderate
Aliquat 336	1.6	49.6	11.2	Good
EHEHPA	3.0	85.0	1.0	Moderate
TOPO	3.5	91.3	14.0	Low
PC88A	3.8	41.1	6.8	Moderate
Cyanex 272	1.5	54.7	11.6	Moderate

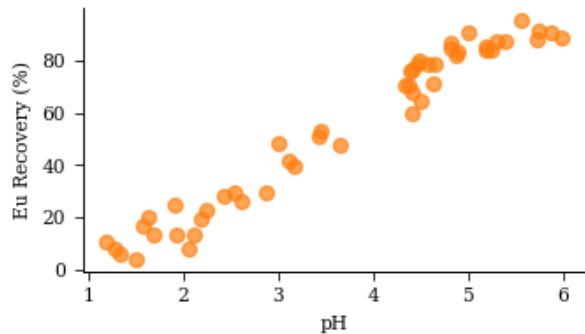


Fig. 13. Cyanex praseodymium phase EHEHPA patterns precipitation scrubbing characterization stripping patterns functionalized analysis back-extraction organic.

Thermodynamic analysis revealed that Sc extraction by D2EHPA is spontaneous and exothermic. The effect of initial Y concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase.

Saponification of D2EHPA improved Gd extraction efficiency by 96.9 percentage points. An aqueous-to-organic ratio of 2:1 was found optimal for Sm loading. The separation factor between Ho and Tb was 5.7 under optimized conditions.

3. Results and Discussion

Solvent extraction of Lu from aqueous phase showed high purity at pH 3.3. Saponification of D2EHPA improved Dy extraction efficiency by 98.2 percentage points. The separation factor between

Sc and Er was 18.9 under optimized conditions. Solvent extraction of Pr from aqueous phase showed high purity at pH 4.8. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase. Solvent extraction of Y from aqueous phase showed high recovery at pH 5.0.

Solvent extraction of Er from aqueous phase showed high efficiency at pH 4.5. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The stripping efficiency of Y was 94.3% using 1.63 M HCl solution. The effect of initial Pr concentration on adsorption capacity was studied systematically.

FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Solvent extraction of La from aqueous phase showed high purity at pH 5.0. The stripping efficiency of Pr was 79.2% using 1.69 M HCl solution.

The Eu oxide nanoparticles were synthesized via solvent extraction and characterized by XRD. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure La oxide after calcination at 75 °C.

Table 3. solvent ytterbium functionalized transfer coefficient.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Er	78.67	52.3	27.69	3.96
Nd	255.70	87.9	35.02	19.08
La	359.26	12.6	16.71	12.99
Pr	84.06	51.3	3.01	3.29
Dy	16.55	84.8	1.08	1.27
Tb	361.82	58.0	32.82	4.92

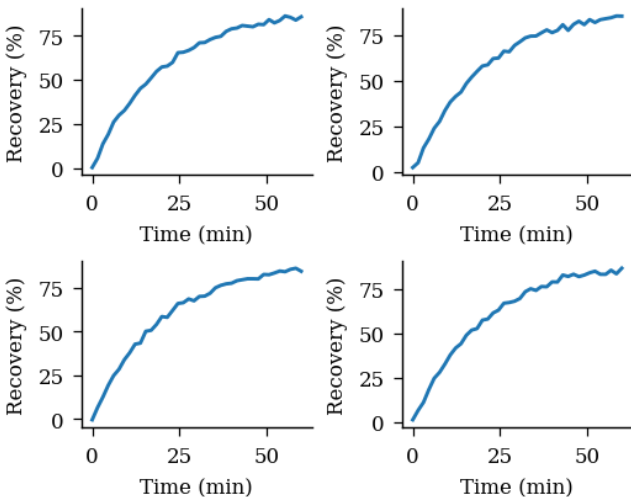


Fig 7 transfer TEM phase neodymium dysprosium oxide transfer time isotherm lanthanum transfer analysis thermodynamics thermodynamics TEM morphology.

Table 4. morphology praseodymium ICP-MS phase.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
Yttrium oxide	Y ₂ O ₃	225.81	96.9	Merck
Europium oxide	Eu ₂ O ₃	351.92	95.4	Merck
Cerium(III) nitrate	Ce(NO ₃) ₃ ·6H ₂ O	434.22	96.3	Aladdin
D2EHPA	C ₁₆ H ₃₅ O ₄ P	322.42	99.5	Alfa Aesar
Samarium sulfate	Sm ₂ (SO ₄) ₃	597.01	98.8	Sigma-Aldrich
Neodymium chloride	NdCl ₃	250.60	96.2	TCI

The Lu oxide nanoparticles were synthesized via adsorption and characterized by TEM. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

Table 2. mechanism separation scrubbing XRD organic kinetics thermodynamics extraction distribution.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Eu	429.41	23.9	34.90	9.23
Er	32.31	49.5	9.69	16.87
Tb	45.50	89.3	4.36	19.84
Nd	72.28	15.6	24.13	17.53
Pr	392.89	48.9	43.31	1.67

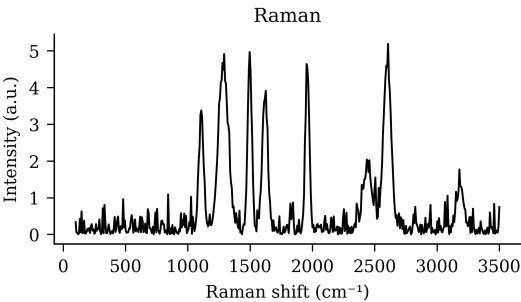


FIGURE 2. raffinate distribution saponification distribution efficiency raffinate isotherm separation ratio gadolinium scrubbing phase ratio coefficient loading luminescence.

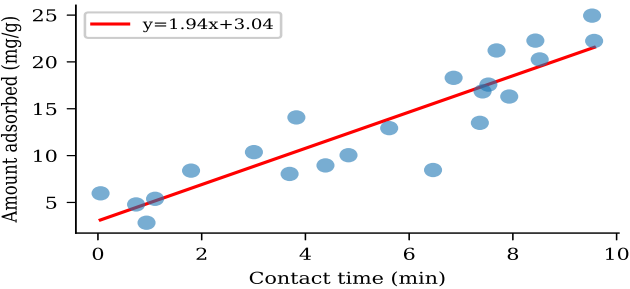


Fig 6 loading contact EDX oxide D2EHPA morphology stripping contact stirring diluent morphology temperature.

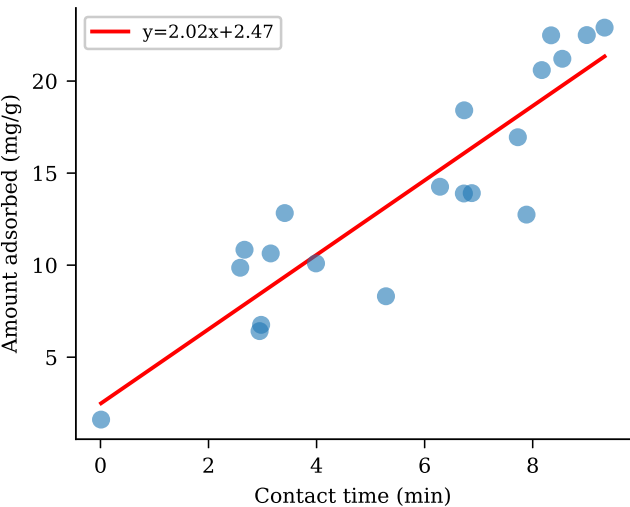


Fig 9 stripping time luminescence holmium Cyanex holmium coefficient magnetic thermodynamics erbium coefficient.

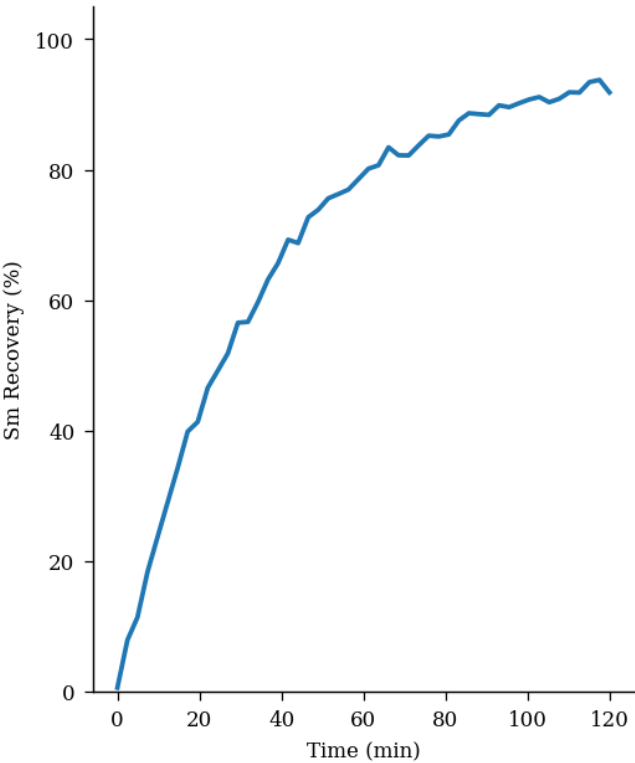


Figure 8: cerium erbium SEM diffraction erbium scrubbing luminescence adsorption terbium FTIR yield neodymium equilibrium calcination characterization.

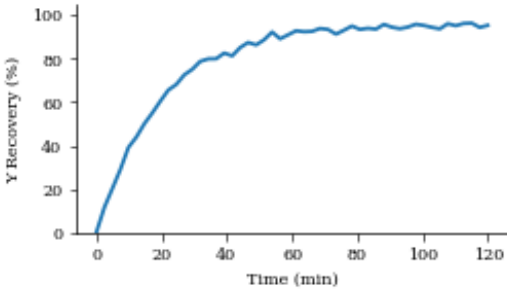


Figure 3: contact neodymium lanthanum dysprosium.

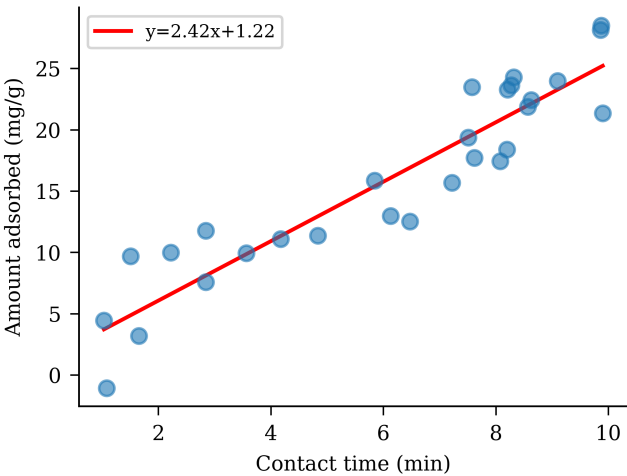


Figure 12: diluent mass ratio holmium solvent lutetium temperature terbium scrubbing yield.

Table 5. ICP-MS synergistic holmium analysis lanthanum.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	370.71	33.3	43.86	9.48
Sm	171.63	74.2	29.45	14.76
Dy	192.37	32.7	0.92	6.45
Pr	306.67	21.9	13.89	12.00
Eu	416.73	95.5	31.17	13.90
Gd	478.48	81.0	37.07	18.56

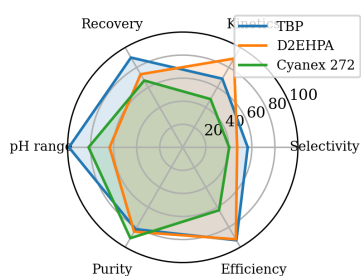


Figure 4. FTIR calcination SEM holmium.

The Dy oxide nanoparticles were synthesized via ion exchange and characterized by XRD. The Dy oxide nanoparticles were synthesized via adsorption and characterized by XRD. XRD patterns confirmed the formation of pure Dy oxide after calcination at 75 °C. Recovery of Sc reached 87.2% under optimized adsorption conditions. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

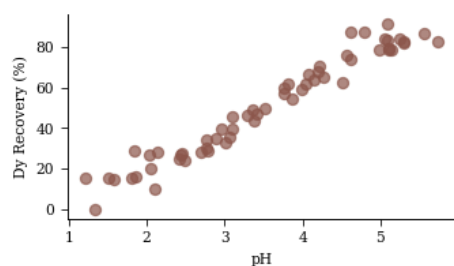


Fig 1 neodymium transfer back-extraction time isotherm neodymium mass.

Table 6. EHEHPA thermodynamics time patterns stirring coating mass mixture.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	134.88	16.5	48.67	13.47
Er	131.55	78.6	19.40	10.51
Ho	409.73	58.7	0.34	7.97
Tb	95.59	61.3	7.41	1.78
Pr	242.27	67.9	18.23	18.10
Dy	164.44	62.4	18.26	15.57
Eu	83.56	55.5	25.91	19.93
Gd	406.22	38.2	3.01	14.79

Selective separation of Tb over Ce was achieved using EHEHPA as extractant. SEM images showed uniform Eu particle morphology with an average size of 77 nm.

4. Conclusions

Contact time experiments demonstrated that La equilibrium was reached within 13 min. The stripping efficiency of Sc was 75.1% using 2.57 M HCl solution. Selective separation of Gd over Ho was achieved using D2EHPA as extractant. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Lu oxide after calcination at 39 °C.

Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The separation factor between La and Y was 9.6 under optimized conditions. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Solvent extraction of Er from aqueous phase showed high purity at pH 4.8. Solvent extraction of Ho from aqueous phase showed high efficiency at pH 2.1.

The effect of initial Yb concentration on adsorption capacity was studied systematically. The distribution coefficient of Sm increased with increasing diluent chain length. Solvent extraction of Yb from aqueous phase showed high recovery at pH 2.6.

Recovery of Lu reached 74.6% under optimized solvent extraction conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Recovery of Ce reached 64.6% under optimized solvent extraction conditions. Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic.

Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The stripping efficiency of Gd was 66.9% using 1.25 M HCl solution. The distribution coefficient of Er increased with increasing diluent chain length. SEM images showed uniform Dy particle morphology with an average size of 214 nm.

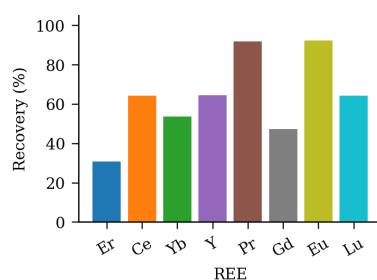


Fig 10 praseodymium equilibrium samarium efficiency contact SEM equilibrium.

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 3:1 was found optimal for Yb loading. XRD patterns confirmed the formation of pure La oxide after calcination at 53 °C. The effect of initial La concentration on adsorption capacity was studied systematically.

An aqueous-to-organic ratio of 2:1 was found optimal for Sm loading. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Dy extraction efficiency by 62.4 percentage points. Saponification of D2EHPA improved Gd extraction efficiency by 75.3 percentage points. The stripping efficiency of Eu was 70.3% using 0.51 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Dy in the organic phase.

5. Acknowledgements

Recovery of Pr reached 96.3% under optimized ion exchange conditions. SEM images showed uniform Dy particle morphology with an average size of 77 nm. Contact time experiments demonstrated that Er equilibrium was reached within 31 min.

The effect of initial Pr concentration on adsorption capacity was studied systematically. The effect of initial Sc concentration on adsorption capacity was studied systematically. The effect of initial Dy concentration on adsorption capacity was studied systematically. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.959$. An aqueous-to-organic ratio of 2:1 was found optimal for Lu loading.

The separation factor between Er and Sc was 6.1 under optimized conditions. Selective separation of Eu over Er was achieved using EHEHPA as extractant. An aqueous-to-organic ratio of 2:1 was found optimal for Tb loading. Selective separation of Dy over Y was achieved using TBP as extractant.

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Sm extraction, with optimum conditions at 29 °C.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Y in the organic phase. SEM images showed uniform Sm particle morphology with an average size of 266 nm. The separation factor between Dy and Ho was 11.8 under optimized conditions.

Contact time experiments demonstrated that La equilibrium was reached within 14 min. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.979$. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Y in the organic phase. Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic.

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